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United States Patent	12408340
Kind Code	B2
Date of Patent	September 02, 2025
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Memory device including a conductive film with interfacial roughness

Abstract

According to one embodiment, a semiconductor device includes a tunnel insulating film, a charge trap film on the tunnel insulating film, and a block insulating film on the charge trap film. The charge trap film is between the tunnel insulating film and the block insulating film. A conductive film is on the block insulating film. The block insulating film is between the charge trap film and the conductive film. The conductive film includes a first metal film adjacent to the block insulating film and a second metal film on the first metal film. The first metal film is between the block insulating film and the second metal film. The first metal film has an interfacial roughness on a side facing the second metal film that is greater than an interfacial roughness on a side facing the block insulating film.

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Appl. No.: 17/681357

Filed: February 25, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230092843 A1	Mar. 23, 2023

Foreign Application Priority Data

JP	2021-154913	Sep. 22, 2021
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Publication Classification

Int. Cl.: H10B43/27 (20230101); H10B41/27 (20230101); H10B41/30 (20230101); H10B41/35 (20230101); H10B43/30 (20230101); H10B43/35 (20230101)

U.S. Cl.:

CPC H10B43/27 (20230201); H10B41/27 (20230201); H10B41/30 (20230201); H10B41/35 (20230201); H10B43/30 (20230201); H10B43/35 (20230201);

Field of Classification Search

CPC: H01L (29/40117); H10B (43/27); H10B (43/35); H10B (41/27); H10B (41/30); H10B (41/35)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2021-154913, filed Sep. 22, 2021, the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments described herein relate generally to a semiconductor device incorporating a three-dimensional memory array or the like.

BACKGROUND

(3) A three-dimensional memory is a known example of a semiconductor device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a semiconductor device according to a first embodiment.

(2) FIG. 2 is a cross-sectional view of a memory cell array.

(3) FIG. 3 is a cross-sectional view taken at a III-III line in FIG. 2.

(4) FIG. 4 is a view showing a more detailed structure example of a conductive film and its surroundings.

(5) FIGS. 5 to 7 are cross-sectional views illustrating aspects related to a method for manufacturing a semiconductor device according to an embodiment.

(6) FIG. 8 depicts a transmission electronic microscope (TEM) image of a conductive film and its surroundings.

(7) FIG. 9 illustrates aspects related to a definition of interface roughness.

DETAILED DESCRIPTION

(8) Embodiments provide a technique for reducing diffusion of oxygen out of a conductive film due to a post-heat load.

(9) In general, according to one embodiment, a semiconductor device includes a tunnel insulating film, a charge trap film on the tunnel insulating film, and a block insulating film on the charge trap film. The charge trap film is between the tunnel insulating film and the block insulating film. A conductive film is on the block insulating film. The block insulating film is between the charge trap film and the conductive film. The conductive film includes a first metal film adjacent to the block insulating film and a second metal film on the first metal film. The first metal film is between the block insulating film and the second metal film. The first metal film has an interfacial roughness on a side facing the second metal film that is greater than an interfacial roughness on a side facing the block insulating film.

(10) Hereinafter, certain example embodiments according to the present disclosure will be described with reference to the drawings. These example embodiments are not intended to limit the present disclosure. In the following description, references to upper and lower directions and the like are relative directions with the surface of a semiconductor substrate on which a semiconductor element is provided being taken as an upper surface. These upper and lower directions are not

necessarily in any particular relationship to gravitational acceleration. The drawings show schematic or conceptual examples, which may differ from actual components in such things as scale, dimensions, relative dimensions, number of components, ratios of parts, and the like. In the specification, elements that are substantially similar to those described in relation to one or more drawings already explained, are given the same reference symbols, and detailed descriptions thereof are appropriately omitted.

First Embodiment

(11) FIG. 1 is a perspective view showing a structure example of a semiconductor device **100a** according to a first embodiment. Herein, a stacking direction of a stacked body **2** is defined as a Z direction. A direction crossing the Z direction is defined as a Y direction. Another direction crossing each of the Z direction and the Y direction is defined as an X direction.

(12) In this first embodiment, the semiconductor device **100a** is a non-volatile memory including a three-dimensional memory cell array in which multiple memory cells are three-dimensionally arranged. For example, the non-volatile memory is a NAND flash memory.

(13) The semiconductor device **100a** includes a base body **1**, a stacked body **2**, and multiple columnar parts CL.

(14) The base body **1** has a substrate **10**, an insulating film **11**, a conductive film **12**, and a semiconductor part **13**. The insulating film **11** is provided on the substrate **10**. The conductive film **12** is provided on the insulating film **11**. The semiconductor part **13** is provided on the conductive film **12**.

(15) The substrate **10** is a semiconductor substrate, for example, a silicon substrate. A surface region of the substrate **10** is provided with, for example, an element isolation area **10i**. The element isolation area **10i** is an insulating region comprising, for example, silicon oxide, and it serves to define an active area AA on the surface region of the substrate **10**. The active area AA is provided with source and drain regions of a transistor Tr or the like. The transistor Tr constitutes a part of a peripheral circuit of the non-volatile memory. The insulating film **11** comprises, for example, silicon oxide, and it insulates the transistor Tr from surrounding elements or the like as necessary. A wiring **11a** is provided in the insulating film **11**. The wiring **11a** is electrically connected to the transistor Tr. The conductive film **12** is a conductive metal, such as tungsten. The semiconductor part **13** is a semiconductor material, such as silicon.

(16) The stacked body **2** is provided above the substrate **10** and is positioned on the semiconductor part **13** in the Z direction. The stacked body **2** includes multiple conductive films **21** and multiple insulating films **22** that are alternately stacked on one another in the Z direction. Thus, the Z direction is the stacking direction of the stacked body **2**. The conductive film **21** comprises, for example, molybdenum. The insulating film **22** comprises, for example, silicon oxide. The insulating film **22** electrically insulates the conductive films **21**, which are adjacent in the Z direction, from one another. The number of stacked conductive films **21** and insulating films **22** can be freely selected. In some examples, insulating film **22** may be, for example, a gap or void, which may sometimes be referred to as an air gap.

(17) In the present non-limiting example, an insulating film **2g** is provided between the stacked body **2** and the semiconductor part **13**. The insulating film **2g** comprises, for example, silicon oxide. The insulating film **2g** may be a high dielectric material having a dielectric constant higher than that of silicon oxide.

(18) The conductive films **21** provide at least one source-side select gate SGS, multiple word lines WL, and at least one drain-side select gate SGD. A source-side select gate SGS is a gate electrode of a source-side select transistor STS. A word line WL is a gate electrode of a memory cell MC. A drain-side select gate SGD is a gate electrode of a drain-side select transistor STD. The source-side select gate SGS is disposed in a lower region of the stacked body **2**. The drain-side select gate SGD is disposed in an upper region of the stacked body **2**. The word lines WL are provided between the source-side select gate SGS and the drain-side select gate SGD. There may be more than one

source-side select gate SGS and the drain-side select gate SGD when more than one source-side select transistor STS and more one drain-side select transistor STD is provided, respectively.

(19) The semiconductor device **100a** includes multiple memory cells MC that are connected in series between the source-side select transistor STS and the drain-side select transistor STD. The structure of the source-side select transistor STS, the multiple memory cells MC, and the drain-side select transistor STD connected in series is referred to as a “memory string” or a “NAND string”. Each memory string is connected to a bit line BL via a contact Cb or the like. The bit lines BL are provided above the stacked body **2** and extend in the Y direction.

(20) FIG. **2** is a cross-sectional view showing an example of a three-dimensional structure of the memory cell array. FIG. **3** is a cross-sectional view taken at III-III line in FIG. **2**.

(21) Each of the multiple columnar parts CL depicted in FIG. **1** is provided in a memory hole MH that extends through the stacked body **2**. The memory hole MH is provided in the Z direction in such a manner as to penetrate through the stacked body **2** from an upper end of the stacked body **2**. A semiconductor body **210**, a memory film **220**, and a core layer **230** are provided inside the memory hole MH. At least some components of the memory film **220**, and the semiconductor body **210** and the core layer **230** extend in the Z direction along the memory hole MH. The semiconductor body **210** can be electrically connected to the semiconductor part **13** and may be referred to as a semiconductor pillar or the like. The columnar parts CL that are arranged in the same row or the like along the Y direction are connected to the same bit line BL via the contacts Cb.

(22) As illustrated in FIG. **3**, the memory hole MH has, for example, a circular or elliptical shape in the X-Y plane. A first block insulating film **21a** that constitutes a part of the memory film **220** is provided between the conductive film **21** and the insulating film **22**. The first block insulating film **21a** is an insulating metal oxide film, such as of aluminum oxide. The first block insulating film **21a** is provided around the conductive film **21**, and it is also provided between the conductive film **21** and the semiconductor body **210** and between the conductive film **21** and the insulating film **22**. The first block insulating film **21a** reduces back-tunneling of electric charges from the conductive film **21** to the memory film **220**.

(23) The semiconductor body **210** has, for example, a cylindrical shape. The semiconductor body **210** is a semiconductor material, such as polysilicon. The semiconductor body **210** serves as a channel region of each of the drain-side select transistor STD, the memory cell MC, and the source-side select transistor STS.

(24) The memory film **220** includes the first block insulating film **21a**, a second block insulating film **221**, a charge trap film **222**, and a tunnel insulating film **223**. At least some components of the memory film **220** are cylindrically formed in such a manner as to extend in the Z direction along the inner wall of the memory hole MH. In addition, the memory film **220** is interposed between the semiconductor body **210** and the conductive film **21** or the insulating film **22**. The memory cells MC have the memory film **220** between the semiconductor body **210** and the conductive films **21** serving as word lines WL. This portion of the memory film **220** between the semiconductor body **210** and the conductive films **21** function as storage regions. The memory cells MC sharing the same semiconductor body **210** are thus arranged along the Z direction one above the other. Each of the semiconductor body **210**, the tunnel insulating film **223**, the charge trap film **222**, and the second block insulating film **221** is deposited along the inner wall of the memory hole MH and extends in the Z direction.

(25) The second block insulating film **221** is provided between the insulating film **22** and the charge trap film **222** and between the conductive film **21** and the charge trap film **222**. The second block insulating film **221** is, for example, silicon oxide. The second block insulating film **221** prevents the charge trap film **222** from being etched when a sacrificial film used in an intermediate manufacturing stage is replaced with the conductive film **21**. The first block insulating film **21a** and the second block insulating film **221** prevent electric charges from leaking from the charge trap

film **222** to the conductive film **21**.

(26) The charge trap film **222** is provided between the second block insulating film **221** and the tunnel insulating film **223**. The charge trap film **222** is, for example, silicon nitride. The charge trap film **222** includes trap sites for trapping electric charges. In the charge trap film **222**, the part (charge trap part) that is interposed between the semiconductor body **210** and a conductive film **21** functioning as a word line WL functions as the storage region of the memory cell MC. The threshold voltage of the memory cell MC varies depending on the amount of electric charges trapped in the charge trap part. With this characteristic, the memory cell MC can store data based on the threshold voltage of the memory cell MC.

(27) The tunnel insulating film **223** is provided between the semiconductor body **210** and the charge trap film **222**. The tunnel insulating film **223** is an insulating material, such as silicon oxide, silicon nitride, or silicon oxynitride film. At the time electrons are injected from the semiconductor body **210** to the charge trap part in a write operation, and at the time holes are injected from the semiconductor body **210** to the charge trap part in an erase operation, these electrons or holes pass or tunnel through a potential barrier of the tunnel insulating film **223**.

(28) In this manner, the tunnel insulating film **223** is provided between the conductive film **21** and the semiconductor body **210**. The charge trap film **222** is interposed between the tunnel insulating film **223** and the conductive film **21**, and the first block insulating film **21a** and the second block insulating film **221** are interposed between the charge trap film **222** and the conductive film **21**. With this structure, the charge trap film **222** captures and releases electric charges from the semiconductor body **210** via the tunnel insulating film **223**. On the other hand, the first block insulating film **21a** and the second block insulating film **221** do not pass electric charges to the conductive film **21** that accumulate in the charge trap film **222**, and they also do not pass electric charges from the conductive film **21** to the charge trap film **222**. Thus, the memory cell MC can store and erase data in the charge trap film **222**.

(29) The core layer **230** fills the inner space of the cylindrical semiconductor body **210**. The core layer **230** has, for example, a columnar shape, and is an insulating material, such as silicon oxide.

(30) The semiconductor device **100a** also includes a semiconductor part **14** (see FIG. 1). The semiconductor part **14** is positioned between the stacked body **2** and the semiconductor part **13**. The semiconductor part **14** is provided between the insulating film **2g** and the insulating film **22** closest to the semiconductor part **13**. The semiconductor part **14** functions as a source-side select gate SGS, for example.

(31) FIG. 4 is a cross-sectional view showing a more detailed structure example of the conductive film **21** and its surroundings. The conductive films **21** that function as a word line WL are provided between mutually adjacent insulating films **22** in the Z direction. The first block insulating film **21a** is provided around the conductive film **21**. In this first embodiment, the conductive film **21** includes a first metal film **211**, a second metal film **212**, and a third metal film **213**.

(32) The first metal film **211** comprises nitrogen (N) and titanium (Ti) or tungsten (W). The first metal film **211** comprises, for example, titanium nitride or tungsten nitride. The first metal film **211** is, for example, titanium nitride or tungsten nitride. The first metal film **211** is provided between the first block insulating film **21a** and the second metal film **212**.

(33) The second metal film **212** comprises tungsten (W). The second metal film **212** is, for example, tungsten (W). The second metal film **212** is provided between the first metal film **211** and the third metal film **213**.

(34) The third metal film **213** comprises molybdenum (Mo). The third metal film **213** is, for example, molybdenum (Mo).

(35) A method of forming the conductive films **21** will be described with reference to FIGS. 5 to 7. As illustrated in FIG. 5, the first block insulating film **21a** is formed by depositing a thin film of Al.sub.2O.sub.3 on an inner wall of the space left by removal of a sacrificial film. The first block insulating film **21a** can be formed by use of a thermal chemical vapor deposition (CVD) method,

an atomic layer deposition (ALD) method, or the like.

(36) Subsequently, as illustrated in FIG. 6, a first metal film **211** is formed. The first metal film **211** is formed on the first block insulating film **21a** by using, for example, ammonia (NH₃) as a reduction gas with TiCl₄, WCl₅, or WCl₄ in a thermal CVD method or an ALD method.

(37) Then, a second metal film **212** is formed, as illustrated in FIG. 7. The second metal film **212** is formed on the first metal film **211** by using, for example, hydrogen (H₂) as a reduction gas with WCl₅ or WOCl₄ in a thermal CVD method or an ALD method. Using a tungsten chloride-type gas, such as WCl₅ or WOCl₄, partially etches the first metal film **211** and increases the interface roughness on the second metal film **212** side of the first metal film **211**. In order to facilitate this roughening of the first metal film **211**, it is desirable to set the deposition temperature of the first metal film **211** to be low and the deposition temperature of the second metal film **212** to be high. It is particularly desirable to set the deposition temperature of the first metal film **211** to be lower than that of the second metal film **212**. In addition, increasing chlorine that is contained in the first metal film **211** enables increasing the interface roughness of the first metal film **211**. In one example, chlorine that is contained in the first metal film **211** can be increased by reducing the amount of NH₃ or increasing the amount of TiCl₄, WCl₅, or WOCl₄, in forming the first metal film **211**.

(38) Subsequently, a third metal film **213** is formed. The third metal film **213** is formed on the second metal film **212** by using H₂ as a reduction gas with MoOCl₄ or MoO₂Cl₂, in a thermal CVD method or an ALD method. Upon forming the third metal film **213**, the conductive film **21** is completed, and the structure as illustrated in FIG. 4 is provided.

(39) FIG. 8 depicts a TEM image of parts of the first block insulating film **21a** (region labeled “AlO_x”), the first metal film **211** (region labeled “TiN”), and the second metal film **212** (region “W”). A variation in interface of each measured position can be calculated, as illustrated in FIG. 9. Specifically, a variation Zn relative to an average height is calculated at each position.

(40) An arithmetic average roughness (Ra value) is calculated by the formula f1 based on the variations Zn.

$$(41) R_a = \frac{1}{N} \cdot \sum_{n=1}^N Z_n \quad (f1)$$

(42) A root mean square roughness (RMS value) is calculated by the formula f2 based on the variations Zn.

$$(43) \text{RMS} = \sqrt{\frac{1}{N} \cdot \sum_{n=1}^N Z_n^2} \quad (f2)$$

(44) As described above, the semiconductor device **100a** of this present embodiment includes the charge trap film **222**, the conductive film **21**, and the first block insulating film **21a** and the second block insulating film **221**, which are provided between the charge trap film **222** and the conductive film **21**. The conductive film **21** is formed as a stacked film of the first metal film **211**, the second metal film **212**, and the third metal film **213**. The interface roughness (interfacial roughness) on the second metal film **212** side of the first metal film **211** is greater than that on the first block insulating film **21a** side (opposite side) of the first metal film **211**. That is, the Ra value at the interface on the second metal film **212** side of the first metal film **211** is greater than that at the interface on the first block insulating film **21a** side of the first metal film **211**. In addition, the RMS value at the interface on the second metal film **212** side of the first metal film **211** is greater than that at the interface on the first block insulating film **21a** side of the first metal film **211**.

(45) This high interface roughness on the second metal film **212** side of the first metal film **211** allows the interface between the first metal film **211** and the second metal film **212** to better trap oxygen that diffuses out from the third metal film **213** due to a post-heat load. Thus, it is possible to reduce deterioration in electric charge retaining characteristics due to oxygen diffusion to the

charge trap film **222**.

(46) The first metal film **211** comprises TiN or WN. The second metal film **212** comprises W. The third metal film **213** comprises Mo.

(47) The Ra value of the interface roughness on the second metal film **212** side of the first metal film **211** is, for example, in a range of 0.2 nm to 1.0 nm, endpoints inclusive. The RMS value of the interface roughness on the second metal film **212** side of the first metal film **211** is, for example, in a range of 0.3 nm to 1.5 nm, endpoints inclusive.

(48) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

Claims

1. A semiconductor device, comprising: a tunnel insulating film; a charge trap film on the tunnel insulating film; a block insulating film on the charge trap film, the charge trap film being between the tunnel insulating film and the block insulating film; and a conductive film on the block insulating film, the block insulating film being between the charge trap film and the conductive film, the conductive film including a first metal film adjacent to the block insulating film and a second metal film on the first metal film, the first metal film being between the block insulating film and the second metal film, wherein the first metal film has an interfacial roughness on a side facing the second metal film that is greater than an interfacial roughness on a side facing the block insulating film, and the first metal film directly contacts the block insulating film.
2. The semiconductor device according to claim 1, wherein the first metal film comprises titanium nitride.
3. The semiconductor device according to claim 1, wherein the first metal film comprises tungsten nitride.
4. The semiconductor device according to claim 1, wherein the first metal film comprises titanium and nitrogen.
5. The semiconductor device according to claim 1, wherein the first metal film comprises tungsten and nitrogen.
6. The semiconductor device according to claim 1, wherein the second metal film comprises tungsten.
7. The semiconductor device according to claim 1, wherein the conductive film further includes a third metal film on the second metal film, the second metal film being between the first metal film and the third metal film.
8. The semiconductor device according to claim 7, wherein the third metal film comprises molybdenum.
9. The semiconductor device according to claim 1, wherein an arithmetic average roughness value of the interface roughness on the second metal film side of the first metal film is in a range of 0.2 nm to 1.0 nm, endpoints inclusive.
10. The semiconductor device according to claim 1, wherein a root mean square roughness value of the interface roughness on the second metal film side of the first metal film is in a range of 0.3 nm to 1.5 nm, endpoints inclusive.
11. The semiconductor device according to claim 1, further comprising: a semiconductor pillar extending lengthwise in a first direction, wherein the tunnel insulating film is on an outer peripheral surface of the semiconductor pillar, the conductive film surrounds the semiconductor pillar in a

plane perpendicular to the first direction, and the block insulating film includes a portion adjacent to the conductive film in the first direction.

12. The semiconductor device according to claim 1, wherein each conductive layer includes a third metal film, and the third metal film directly contacts the second metal film.
 13. A memory device, comprising: a plurality of conductive films spaced from one another in a first direction; and a plurality of semiconductor pillars extending lengthwise in the first direction through the plurality of conductive films, each semiconductor pillar having a tunnel insulating film on an outer peripheral surface thereof, a charge trap film on the tunnel insulating film, and a block insulating film on the charge trap film with the charge trap film being between the tunnel insulating film and the block insulating film, wherein each conductive film surrounds the semiconductor pillars in a plane perpendicular to the first direction and contacts the block insulating film on the semiconductor pillars, each conductive film includes a first metal film adjacent to the block insulating film and a second metal film on the first metal film, the first metal film being between the block insulating film and the second metal film, the first metal film has an interfacial roughness on a side facing the second metal film that is greater than an interfacial roughness on a side facing the block insulating film, and the first metal film comprises either tungsten nitride or titanium and nitrogen.
 14. The memory device according to claim 13, wherein each conductive film further includes a third metal film on the second metal film, the second metal film is between the first metal film and the third metal film, and the third metal film is molybdenum.
 15. The memory device according to claim 13, wherein an arithmetic average roughness value of the interface roughness on the second metal film side of the first metal film is in a range of 0.2 nm to 1.0 nm, endpoints inclusive.
 16. The memory device according to claim 13, wherein a root mean square roughness value of the interface roughness on the second metal film side of the first metal film is in a range of 0.3 nm to 1.5 nm, endpoints inclusive.
 17. A semiconductor device, comprising: a stacked body comprising alternating insulating layers and conductive layers stacked in a first direction; and a plurality of memory pillars extending through stacked body in the first direction, wherein each conductive layer has an outer film layer facing the insulating layers and the memory pillars and a first inner film layer adjacent to the outer film layer, the outer film layer has an interfacial roughness on a side facing the first inner film layer that is greater than an interfacial roughness on a side facing away from the first inner film layer, each conductive layer includes a second inner film layer, and the first inner layer is between the second inner layer and the outer film layer.
 18. The semiconductor device according to claim 17, outer film layer comprises tungsten nitride and the first inner film layer is metallic tungsten.
 19. The semiconductor device according to claim 18, wherein the second inner film layer is molybdenum.
 20. The semiconductor device according to claim 17, wherein the second inner film layer comprises molybdenum.
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